

PROFESSIONAL ENVIRONMENTS 1
(CI4450_ALL_TY13_24)

TB1 COURSEWORK 1

NAVIGATING ARTIFICIAL INTELLIGENCE IN THE
WORKPLACE

TECHNICAL REPORT

Alpha 6		6 th , Dec 2024
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1. Introduction

1.1. History of Artificial Intelligence (AI)

The concept of machines acting, thinking, and feeling like humans, can be dated as far back as the 1700s. In 1763, Thomas Bayes' research on the "probability of the validity of a proposition, now called Bayesian estimation", started the basis for machine learning algorithms, used in AI systems today. (Weal, 2024)

It was only in 1950 that the pioneer Alan Turing, started the research known as the "Turing Test", that laid the foundation for AI systems as we know it today. (Lawrence Livermore National Laboratory, 2022)

1.2. Types of Artificial Intelligence

Artificial Intelligence (AI) is a technology that enables systems to perform tasks, that otherwise would only be possible with human intervention.

There are 4 main types of AI based on functionality, recognized today, "**Reactive Machine**" and "**Limited Memory**", the most used nowadays, "**Theory of Mind**" and "**Self-Aware**", being mostly theoretical. (Team, 2024)

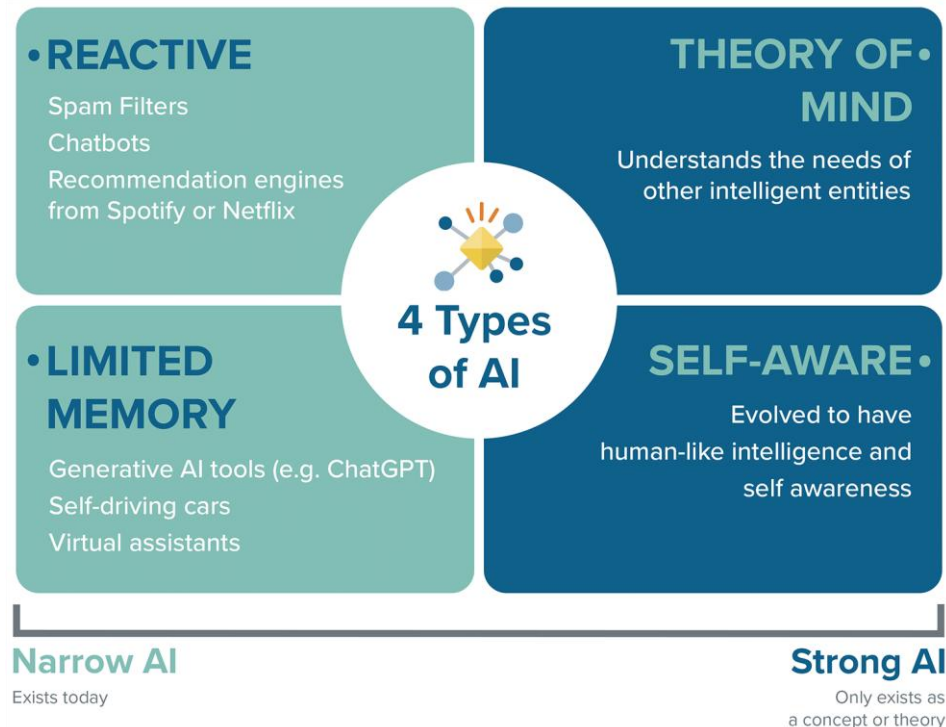


Figure 1 (Huang, 2023)

1.3. Artificial Intelligence Technologies, Products and Services

Developments in the field are mainly due to key technologies like “**Machine Learning**”, “**Natural Language Processing**”, “**Computer Vision**”, “**Deep Learning**”, “**Generative Models**”, and “**Expert Systems**”. These examples of technologies enable AI systems to learn from experience, recognize text and spoken words, understand and extract information from visual inputs, process data like humans and react correspondingly, generate new content from input data that looks like the original content, and use a collection of facts and heuristics to give reliable complex decisions. (Takyar, 2023)

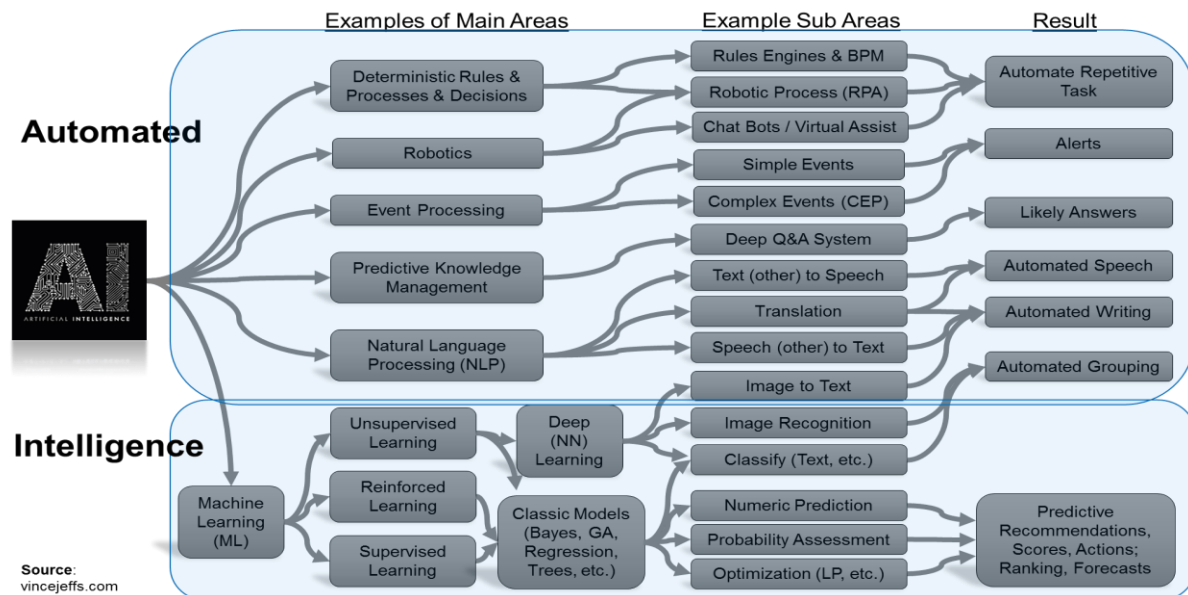


Figure 2 (Marketing Technology, 2024)

1.4. Societal Integration

AI products and services can be used on several devices including laptops, phones, TVs, and even in cars, making it easily accessible to everyone, everywhere. The ease of access coupled with the extended functionalities made it easy for AI to be introduced into society.



Figure 3 (AIPRM, 2024)

2. Impact of Artificial Intelligence on Education

2.1. Stakeholders

The introduction of AI into the educational systems transformed learning and administrative operations. It impacts main stakeholders like students, teachers, and institutions, offering tools that automate, expand, adapt, and manage tasks, giving stakeholders time to focus on more important areas.

STAKEHOLDER ANALYSIS			
Stakeholders	Interests	Responsibilities	SOCIETY
Students The main beneficiaries	- Offers tools to enhance learning experiences. - Help develop skills necessary to succeed.	A responsible and ethical use of products and services.	
Teachers The forefront of AI implementation in the classrooms	- Help teachers identify students requiring support. - Supply tools to manage classroom material.	- Use of best approach to better equip students. - Willingness and readiness to accept changes.	
Institutions Create policies and the foundation of educational curriculums	Economic growth by providing skilled graduates.	- Provide training to teachers. - Fund infrastructures. - Constant research to provide sustainable and innovative solutions.	

2.2. Technologies, Products, and Services on Education

AI-powered products and services became an essential part of modern education, aiding stakeholders to achieve optimal outcomes.

- **Speech Recognition Tools:** Duolingo helps students improve their language skills.
- **Assistive Technology:** Microsoft's Immersive Reader provides tools to help students with learning disabilities and visual impairments.
- **Skills Mapping Platforms:** LinkedIn and Coursera assess students' abilities to recommend career paths or educational resources for skill development.
- **Learning Management Systems Analytics:** Canvas and Blackboard provide vital information to teachers, to identify students who require extra support.
- **Automated Grading Systems:** Turnitin and Gradescope grade written responses, detect plagiarism, and assess handwritten work.

These are just some examples of how applications are used, in conjunction with technologies, to create adaptive and immersive educational environments.

2.3. Positive Impact

AI significantly improved the educational system, enabling stakeholders to create more personalised, efficient and productive learning environments.

- **Personalised Learning:** Improves accessibility, especially for students with disabilities or language barriers.
- **Feedback:** Enhances student literacy and performance, identifying areas that require improvement.
- **Management Tools:** Helps teachers with classroom resources while giving them more time to focus on students.

2.4. Negative Impact and Hindsight View

Challenges are also present that could be predicted and mitigated.

- **Bias and Ethical:** AI algorithms can perpetuate biases in existing data, and lead to discrimination. Regular monitoring and evaluation of AI systems and databases can identify and correct any potential biases. Training stakeholders to understand AI processes reduces the risk of misuse.
- **Privacy Concerns:** Information accessed by AI services, and unclear system policies can lead to privacy issues. Transparency on AI-gathered information and opt-in policies for data collected ensures consent and promotes trust.
- **Personal Development:** Increased use of AI tools can cause dependency and decrease human connections, reducing students' creativity, ability to think critically, and solve problems independently, and may lead to problems in emotional development. Education institutions should design curricula that include group activities, debates, and collaborative projects to promote social interaction, critical thinking, and problem-solving skills, whilst contributing to emotional development.

Integration of AI systems into education directly affects stakeholders, requiring a delicate balance to maintain positive factors while trying to reduce negative aspects.

3. Impact of Artificial Intelligence on the Workplace

3.1. Stakeholders

AI integration is global, making workplaces more competitive by increasing efficiency and responsiveness. The impact of AI in the workplace involves a diverse range of stakeholders, we will focus on employees, employers, and customers, as they are the most directly affected.

STAKEHOLDER ANALYSIS			
Stakeholders	Interests	Responsibilities	SOCIETY
Customers The most affected	- Products that enhance customer services. - Personalisation of the shopping experience.	Stay informed about data usage and privacy rights.	
Employees Collaborate with AI tools	- Time saved and increased efficiency and services quality. - Encourages upskilling. - Enhances creativity.	- Actively participate in AI training to be up-to-date. - Responsible use. - Address potential biases. - Ensure contributions align with business goals.	
Businesses Responsible for an effective integration	- Reduce operational costs. - Streamline workflows to allocate resources more effectively. - Advanced analytics for better decisions.	- Address ethical concerns. - Invest in reskilling and upskilling of employees. - Foster trust and transparency about AI uses.	

As AI continues to interact with people and businesses, it is important to note that, as mentioned by Rupert Younger, Director of Oxford University Centre for Corporate Reputation, “stakeholder engagement lies at the core of corporate strategy. This article argues that AI – particularly generative AI – has now become an important tool for organisations and that its ability to create information and shape opinions requires it to be conceptualised as a stakeholder rather than as a technology.” (Saïd Business School, 2024)

3.2. Industries

This integration is not only in sectors like technology, business, and education, but is extending its transformative potential to a wide range of industries. From healthcare to finance, logistics to entertainment, AI is reshaping how businesses operate, make decisions, and deliver value.

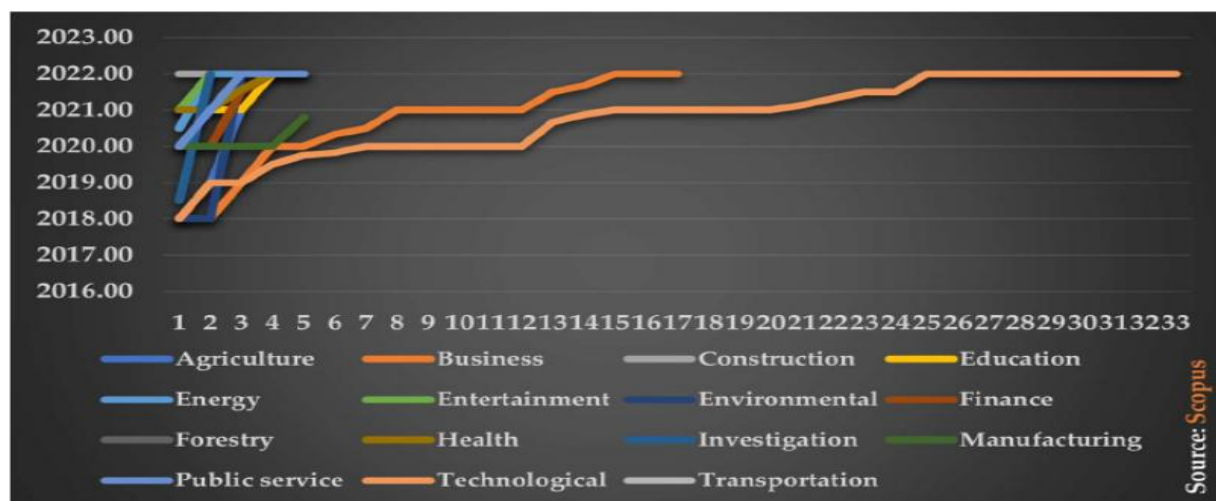


Figure 4 (Espina-Romero et al., 2023)

3.3. Products and Services in the Workplace

AI applications and services include a range of functionalities that automates, analyses, predicts, tracks, and assists, between many others.

- **Automation of Routine Tasks:** Automates tasks like data entry, invoice processing, report generation, and customer service interactions.
- **Data Analysis:** Uses historical data for inventory management and strategy planning, converting raw data into understandable visualisations.
- **Human Resources Management:** Assists with tasks like recruiting, and monitors employee engagement.
- **Customer Relations:** Analyses customer behaviour to identify areas for improvement.
- **Cybersecurity:** Detects unusual patterns to help identify cyber threats quickly and assists with verification checks.
- **Inventory Management:** Predicts demand and analyses various factors for route optimization.
- **Project Management:** Aids task scheduling and communication between global teams.
- **Learning:** Tracks employee performance and personalises training accordingly.

These are just some examples of where AI is used in the workplace, bringing opportunities for stakeholders to optimize their work in all industries.

3.4. Positive Impact

AI implementation in the workplace has a huge potential, enhancing productivity, decision-making, and efficiency in workplaces within various industries.

- **Productivity Tools:** AI services boost productivity, efficiency, and customer satisfaction.
- **Predicting Services:** Reduces waiting times and maintenance costs, and improves administrative and relations processes.
- **Managing Systems:** Helps with repetitive tasks, legal procedures, and makes operations more efficient.

3.5. Negative Impact and Hindsight View

AI also brings many concerns that can be predicted and avoided, or mitigated if already in place.

- **Legal and Ethical Concerns:** Algorithms can lead to discrimination, particularly on hiring, promotions, and performance reports. Information generated might contain personal or sensitive data. Organizations should review systems regularly to avoid bias discrimination, be transparent about policies, and comply with data protection laws.
- **Job Displacement:** AI integration into work operations could lead to the replacement of the current workforce. Understanding how AI complements workplaces, rather than replacing them, along with training programs might defer any job displacement concerns, and create new jobs.

4. Impact on Personal Career Journeys and Workplaces

4.1. Future Workplaces

AI technology landscape is a rapidly changing field, adding new types of skills involved, and creating new paths available. Projects will see improved quality and delivery times through automation, and new job roles might emerge to address the changes. However, we might see the automation of basic jobs, not necessarily cutting the number of available jobs, but most probably changing how these jobs are performed. (Chada, 2024)

This offers students and professionals a unique opportunity to push the boundaries in software development, data science, cybersecurity and more. Opportunities may include expanded roles for data analysts, cloud engineers, software developers, machine learning specialists and more, reflecting the growing demand for diverse expertise in AI-driven industries. (Online, 2024)

Continuous upskilling will allow professionals to specialise in AI subfields, expanding their expertise across different domains. This change requires specific skills, emphasising the importance of taking into consideration the needs of businesses.

4.2. Future Skills and Use Scenarios

Universities worldwide are recognising the importance of equipping students with future skills, especially for AI-driven workplaces, and are designing interdisciplinary courses that combine AI with subjects like business, humanities and the arts, ensuring that future graduates will have an opportunity to adapt and succeed alongside AI systems. According to the Kingston University report, **“Problem-Solving”, “Communication Skills”, “Digital Skills”, “Critical Thinking”** and **“Analytical Skills”** are among the top five skills most valued by businesses. (Future Skills The Kingston Approach, n.d.)

“Problem-Solving”: Encouraging students to work and solve challenges on real-world problems, develop interdisciplinary skills, enabling students to apply AI solutions to dynamic workplaces like healthcare, finance, or logistics.

“Communication Skills”: Hybrid learning environments and project presentations prepare students for roles like AI experts or business strategists, allowing them to communicate complex AI concepts effectively to any kind of audience.

“Digital Skills”: Courses in AI, machine learning and big data give students the hands-on training necessary to successfully operate AI tools and develop new algorithms.

“Critical Thinking”: Combining humanities with AI and encouraging students to evaluate ethical AI dilemmas, provides the basics necessary to assess AI systems for unintended ethical issues and to properly evaluate organisational goals and societal needs.

“Analytical Skills”: Training in programming languages and making students analyse AI models allows them to interpret AI-generated data, refine algorithms, and fine-tune them for scalability and accuracy.

This empowers professionals and new graduates with the set of skills necessary to adapt and excel in AI-driven workplaces.

5. Impact of Artificial Intelligence PESTLE Analysis

AI has a profound impact across all domains, and while it brings innovation and growth opportunities, it also introduces concerns.

To fully understand AI's impact globally, we need to analyse its effects from a Political, Economic, Social, Technological, Legal and Environmental perspective.

Political	• Regulation: Governments are developing policies to regulate AI, focusing ethical use, privacy and data protection. • Strategies: Countries like USA, China, and the EU are investing in AI for economic and military purposes. • Competition: Geopolitical competition potentially leading to innovation hubs or technology monopolies.
Economic	• Economic Growth: Automation of tasks increase productivity, leading to job displacement in some sectors, but simultaneously creating specialised jobs. • Cost Efficiency: Businesses can reduce costs with AI by improving operations and automation of tasks. • Market Disruption: More traditional industries suffer with AI-driven startups that introduce innovative solutions.
Social	• Workforce: Constant upskilling and reskilling creates challenges for people who lack access to resources. • Accessibility: Improves accessibility for people with disabilities and language barriers. • Ethical: Issues like algorithmic bias, data privacy and misinformation raises concerns.
Technological	• Advancements: Technologies enable applications in areas like healthcare, autonomous vehicles, and robotics. • Big Data and Cloud: AI requires vast amounts of data and computational power, driving growth in these areas. • Cybersecurity: Strengthens cybersecurity, but also introduces the risks of AI-driven attacks.
Legal	• Data Protection: Regulations like GDPR and CCPA enforce guidelines on data collection. • Property Rights: Ownership of AI-generated content and use of copyright material by AI algorithms remains a concern. • Accountability: Determining liability in AI-made decisions poses legal challenges.
Environmental	• Energy: Use of AI systems demands significant computational power, increasing carbon emissions. • Sustainability: AI aids the environment by monitoring creating smart energy grids and optimising resources. • E-Waste: Rapid technological advancements promotes upgrades of systems increasing electronic waste.

It is worth adding that governments already recognise the importance of AI integration into the educational system. Research and investment have already started with the UK government investing £2 million in AI tools. (mediaofficer, 2023)

In the workplace, this integration is also seen as vital but predicts 3 million job losses in the process, remaining optimistic with predictions that will open job opportunities, boosting productivity and economic growth. (Sharps *et al.*, 2024)

AI implementation comes with high costs involved, probably leading to the straining of organisations' budgets and resources, but with initial funding, offers better future prospects.

Advancements of AI introduce valuable products and services, that help stakeholders in all domains. This ensures that AI research will intensify.

6. Conclusion

In conclusion, AI has evolved significantly from its initial state. It transformed the way we learn, communicate, work, and live. The full impact is still not fully understood, but we can say for sure that if done right, it holds huge potential for us.

“Artificial intelligence is here today. Its impact is already being felt in the economy, in public life and in day-to-day experiences. Although it's still the very early chapters of the story, the general-purpose nature of the technology and its potential to be adopted quickly make AI hugely consequential.” (Sharps *et al.*, 2024)

In this early chapter that we now find ourselves, the choices we make today will define the trajectory of the technology. Only by fostering innovation while safeguarding human values, we have the opportunity to harness AI's full potential, not as a tool but as a force for shared prosperity.

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